

One idea from the full audit: set the forecast's level with statistics, not with the LLM

GEPA forecaster project · audit of 2026-09-02 (13 read-everything dossiers, 5 independently verified re-analyses, 1 pre-registered fresh run) · \$8.30 of the \$100 budget used

Setup, so every number below is readable: we forecast the probability that each of 11 AI models solves each cyber benchmark task; a *cell* = one (model, task) pair; Brier scores the forecast against the observed pass/fail, lower is better. Sets: *eval* = 84 cells the search optimized on; *sealed* = 230 held-out cells from the same 5 training benchmarks; *reserved test* = 1,033 cells from the 2 held-out benchmarks (CVEBench, CyberGym). GEPA is the automated prompt optimizer we ran; the *seed* is its hand-written starting prompt.

The idea, in one breath. Let simple statistics set each forecast's *level* — that is the part that transfers — keep the LLM at most for *ordering* tasks within a level, and *audit the training data first*. Concretely:

1. **Anchor.** A 12-parameter curve — one ability number per model, one shared “success falls with human solve time” slope — fit *only* on the 5 training benchmarks, sets the level. It beats the seed on eval, sealed, and the reserved test.
2. **Guardrail (one prompt line).** Tasks are binned by human solve time, and every prompt already prints the model's per-bin pass rates; cap the forecast at the printed rate of the adjacent easier bin. This improves the seed on sealed (0.131→0.111) and on the reserved test (0.163→0.156, $t = -3.4$).
3. **Audit.** Refit the curve leaving out one training benchmark at a time: this measures our own data's bias. It exposed InterCode-CTF as contamination-inflated (its solve rate is flat at 0.88–1.00 regardless of difficulty). With that one benchmark removed from the fit, the curve's level matched the reserved test's base rate to 0.001 (0.241 vs. 0.240) — using zero labels from the test benchmarks.

Brier	reserved test (labels already examined offline)	fresh check: 297 never-prompted cells
seed prompt	0.163	0.190
the 7 GEPA-optimized prompts	best ties the seed, worst 0.230	—
curve, audited (zero test labels)	0.147	0.155 (−0.035 vs. seed, $t = -2.6$)
blend 0.64 curve + 0.36 seed, log-odds	0.140–0.145	0.159 (−0.031 vs. seed, $t = -3.6$)
per-(model, bin) table fit on the test labels	0.149	—

Blend weights were chosen on training benchmarks only and frozen before the fresh run; its 0.140–0.145 range spans the audited/unaudited fit. It *matches* the label-fit table (a coarse reference, not an upper bound; the difference is within noise, so “matches” is the defensible word). The fresh check (pre-registered, `gepa runs/holdout27_prereg.md`): 27 tasks from the same two held-out benchmarks — 7 CVEBench + 20 CyberGym excluded from the original test only for lacking curated difficulty labels, never prompted before, ~\$8. Both benchmarks' *offsets* (the one number you would add to the curve's log-odds to fit that benchmark: +0.04, −0.08) landed inside the ± 0.40 spread predicted from the training benchmarks alone, and the unaudited curve overshot in the predicted direction (mean forecast 0.401 vs. realized solve rate 0.320). Note the blend trails the curve on the fresh cells (+0.003, not significant) — see limits.

Why GEPA inverted (diagnosis, not blame). Its acceptance rule never scored a candidate on a benchmark held out from the search, so its edits distilled training statistics into the prompt's confidence *level* — the one component that does not transfer. The optimized prompts' *rankings* of cells transferred fine; every measured harm was a level error (across all 8 measured prompts, higher mean forecast $\Rightarrow$ worse test Brier, rank correlation 1.00).

Honest limits. Two held-out benchmarks only, and 80 of the 94 test tasks are CyberGym. The left column was computed after those labels had been examined — that is why we bought the fresh check. “The LLM still adds ordering value on top of the curve” is supported on the left column ($t = -2.3$) but did *not* appear on the fresh cells (+0.003, n.s., ~13% power): treat it as an open question, not a result.

If the team wants the follow-up. (a) One GEPA rerun (~\$150) whose acceptance rule holds out one training benchmark per fold — you get what you gate on — promotion bar: beat the frozen blend. (b) In deployment, once ~20 real outcomes from a new benchmark resolve, one global recalibration parameter should close most of the remaining gap (offline replay: robust from $k=20$; standard practice in operational forecasting).

Every number above was reproduced by at least two independent derivations from the raw logs. Trail: `gepa FORECASTER_GEPA_RUNS.md` (holdout row), `runs/holdout27_prereg.md` (frozen coefficients, committed before launch), results branch (raw cells); errata from the audit appended to `sweep_and_test_2026-09-01.md`. Preliminary throughout: one dataset, one harness, one 11-model panel.